Figure 4. Larger cells maintain heading longer under fluctuating input

Larger native rings retain heading longer under fluctuating input, with the largest size exponents in the weak-input regime. The baseline examples distinguish geometric persistence from a size-dependent change in physical dynamics.

A, Synthetic activity kymographs for a single drifting peak, same-rate random relocations and relocations whose rate scales with circumference, at C20 and C60. Position is relative arc length; grayscale denotes activity from 0 (white) to 1 (black). Time increases upward. These pedagogical baselines have no biochemical kinetics.

B, Heading correlation in the same three baselines, from 1000 seeds; solid and dashed lines indicate C20 and C60, dotted lines the analytic curves. A fixed physical drift diffusivity gives k proportional to C^{-2} ($\alpha = 2$); size-independent jumps give $\alpha = 0$, and jumps per unit perimeter give $\alpha = -1$. The same-rate curves coincide.

C, Native-model heading correlation at common weak f = 0.005 and strong f = 0.3, for C20, C40 and C60 (Spring 1, 2 and 3). Size is a light-to-dark grayscale ramp within each facet. Shading is the 95% confidence interval from 2000 seed-bootstrap resamples of the complete retained native histories, recomputed on a dense early-lag grid. Weak-input facets show 0–320τ; strong-input facets show 0–12τ, Goryachev 0–80τ, to resolve their different correlation scales. $\tau = 50$ native time units, Spring 5. Heading correlation is the mean cosine of the change in arg(P), where P is the first Fourier activity moment divided by total activity. These retained round6 estimates require $|P| \geq 0.1$ at both lag endpoints; their coverage is in condition_lag_metrics.csv. They differ from the ungated distributions in Figure5D.

D, Native Wave-pinning examples at f = 0.005 and 0.3 and C20, C40 and C60, selected as replicate 0 without a seed search. Every panel shows the entire retained post-burn-in interval 16000–32000, on relative arc position. Activity scales are common across sizes at each input; the color bars state the separate weak- and strong-input ranges. Weak input mainly shifts a coherent domain; strong input produces a changing patch field. The kymographs are illustrative, not an event classification.

E, Local α, the negative fitted slope of log heading-decorrelation rate against log circumference in adjacent-size neighborhoods, against mean input f. Lines are size neighborhoods centered on C20, C30, C40, C50 and C60 (Spring 1, 1.5, 2, 2.5 and 3); light to dark follows size. Vertical bars are seed-bootstrap 95% CIs. Missing estimates interrupt curves. Horizontal guides are $\alpha = 2$, 0 and -1. Native f multiplies reference input 0.1, 1, 4 and 6.4 in Wave-pinning, Holmes, Spring and Goryachev respectively; doses and clocks are not calibrated across models.

All simulations reuse the complete, frozen round6 compact campaign (96–384 independent seeds per condition). Local kinetics, basal preparations and solver outputs are unchanged. α and activity-heading persistence are model observables, distinct from migration persistence or fitted OU velocity-memory parameters.